

Simultaneous B_1^+ and T_1 estimation in multi-pathway multi-echo (MPME) imaging: forward model and fundamental limits

Sarang Joshi

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Abstract

We derive the steady-state signal model of the multi-pathway multi-echo (MPME) sequence of Cheng et al. [1] and analyse the joint estimation of the transmit field B_1^+ and the longitudinal relaxation time T_1 through the Fisher information and the Cramér–Rao lower bound (CRLB). The central difficulty is a property of the *forward model*: for small flip angles and $T_R \ll T_1$ every pathway amplitude is invariant, to leading order, under $(M_0, B_1^+, T_1) \mapsto (M_0/k, kB_1^+, T_1/k^2)$. Any protocol whose flip angles are all small therefore has a nearly singular information matrix, and *no* estimator can separate B_1^+ from T_1 . The near-360° second pulse of the published protocol breaks the invariance through the 2π wrap of the rotation, reducing the condition number from 5×10^5 to 173. What remains is a finite bound ($\approx 40 \sigma/M_0$ relative standard deviation for T_1) set by the data. The published *analytic inverse* adds a further, estimator-specific loss: in Monte Carlo its T_1 spread is 3.1–3.3 times the bound at high SNR and 7.6 times at moderate SNR, where 2% of voxels have no valid T_1 . A joint maximum-likelihood fit of the same model attains the bound at every SNR tested.

How to read this note

Section 1 builds the forward model from the configuration-state (extended phase graph) description up to the measured voxel signal. Section 2 sets up the estimation-theoretic tools. Section 3 proves the scaling invariance that underlies the B_1^+/T_1 problem and explains how the published protocol escapes it. Section 4 gives the bounds and the conditioning of the information matrix. Section 5 answers the question *is the limitation in the forward model or in the inverse?* by comparing the analytic reconstruction and a maximum-likelihood fit against the bound. Table 4 summarises the answer.

All numbers are reproducible with the code in the accompanying repository (Appendix A).

1 Forward model

1.1 Sequence

The MPME sequence is a 3D gradient-echo sequence without RF spoiling, with a constant RF phase φ and repetition time T_R . Only the readout gradient G_x is unbalanced; its net area per repetition, $A = \int_0^{T_R} G_x dt$, dephases the magnetisation by many cycles across a voxel (the *fully dephased* regime). Two acquisitions $i = 1, 2$ use the same gradient waveform with nominal flip

angles $\alpha_1^{\text{nom}} = 15^\circ$ and $\alpha_2^{\text{nom}} = 330^\circ$ and $T_R = 25$ ms [1]. The actual flip angles are

$$\alpha_i = B_1^+ \alpha_i^{\text{nom}}, \quad (1)$$

so B_1^+ scales all pulses by the same factor.

Each of three readout windows of alternating polarity sweeps the cumulative gradient moment through three adjacent integer multiples of A (lobe areas $+1.5 : -3 : +3 : -3 : +0.5$ in units of $-A$), sampling three *pathways* (a two-pathway predecessor is described in [2]) $k \in \{+1, 0, -1\}$ spaced by $1/\text{BW} \approx 2$ ms. A state of order k created by an RF pulse is refocused when the moment accumulated since the pulse equals $-kA$. This gives each pathway three echo times $t_{ijk} \in (0, T_R)$ (in ms, assuming 4.5 ms from the pulse to the first echo): $k = +1$: 4.5, 14.5, 16.5; $k = 0$: 6.5, 12.5, 18.5; $k = -1$: 8.5, 10.5, 20.5.

1.2 Configuration states

Let $M^+ = M_x + iM_y$ and let $\psi \in [0, 2\pi)$ be the intravoxel phase imparted by one repetition of the unbalanced gradient. Expand

$$M^+(\psi) = \sum_k F_k e^{ik\psi}, \quad M_z(\psi) = \sum_k Z_k e^{ik\psi}, \quad Z_{-k} = \overline{Z_k}. \quad (2)$$

One repetition of the operators acts as follows [4, 3].

RF pulse (flip α , phase φ ; rotation $R = R_z(\varphi)R_x(\alpha)R_z(-\varphi)$):

$$F'_k = \cos^2 \frac{\alpha}{2} F_k + e^{2i\varphi} \sin^2 \frac{\alpha}{2} \overline{F_{-k}} - ie^{i\varphi} \sin \alpha Z_k, \quad (3)$$

$$Z'_k = -\frac{i}{2} e^{-i\varphi} \sin \alpha F_k + \frac{i}{2} e^{i\varphi} \sin \alpha \overline{F_{-k}} + \cos \alpha Z_k. \quad (4)$$

Relaxation over T_R , with $E_1 = e^{-T_R/T_1}$ and $E_2 = e^{-T_R/T_2}$:

$$F_k \rightarrow E_2 F_k, \quad Z_k \rightarrow E_1 Z_k \ (k \neq 0), \quad Z_0 \rightarrow E_1 Z_0 + M_0(1 - E_1). \quad (5)$$

Gradient $F_k \rightarrow F_{k+1}$. Free precession by $\Delta\omega$ for a time t multiplies M^+ by $e^{i\Delta\omega t}$ (same sense as the gradient twist).

1.3 Steady state

Write $a_k(\alpha, T_1, T_2, T_R)$ for the steady-state amplitude of F_k immediately after the RF pulse, per unit M_0 . It can be computed exactly without truncating the state space: an isochromat with total precession $\beta = \psi + \Delta\omega T_R$ per repetition has the post-pulse steady state

$$(I - R R_z(\beta) \text{diag}(E_2, E_2, E_1)) \mathbf{M}^+ = (1 - E_1) R \hat{\mathbf{z}}, \quad (6)$$

and F_k is the k -th Fourier coefficient of $M^+(\psi)$ over ψ .

Proposition 1 (Off-resonance only adds phase). *In the fully dephased regime,*

$$a_k(\Delta\omega) = a_k(0) e^{ik\Delta\omega T_R};$$

in particular $|a_k|$ does not depend on $\Delta\omega$.

Proof. By (6) the steady state depends on ψ and $\Delta\omega$ only through $\beta = \psi + \Delta\omega T_R$, so $M_{\Delta\omega}^+(\psi) = M_0^+(\psi + \Delta\omega T_R)$. Shifting a periodic function by θ multiplies its k -th Fourier coefficient by $e^{ik\theta}$. $\square$

For the two lowest pathways the steady-state magnitudes have closed forms (unbalanced FISP and PSIF), which we verified against (6) to 10^{-6} :

$$\begin{aligned} |a_0| &= \left| \tan \frac{\alpha}{2} \left[1 - \frac{(E_1 - \cos \alpha)(1 - E_2^2)}{r} \right] \right|, \\ |a_{-1}| &= \left| \frac{\tan \frac{\alpha}{2}}{E_2} \left[1 - \frac{(1 - E_1 \cos \alpha)(1 - E_2^2)}{r} \right] \right|, \\ r^2 &= p^2 - q^2, \quad p = 1 - E_1 \cos \alpha - E_2^2(E_1 - \cos \alpha), \quad q = E_2(1 - E_1)(1 + \cos \alpha). \end{aligned} \quad (7)$$

The PSIF expression in the literature is the echo amplitude at $t = T_R$, hence the factor $1/E_2$ for the post-pulse state.

1.4 Echo evolution and the voxel signal

Pathway k read at time t after the pulse carries, for one isochromat, the off-resonance phase $\Delta\omega(t + kT_R)$ (Proposition 1 plus precession since the pulse). Averaging over a Lorentzian intravoxel frequency distribution of half-width R'_2 , $\langle e^{i\delta\tau} \rangle = e^{-R'_2|\tau|}$, gives the signal of scan i , echo j , pathway k :

$$S_{ijk} = M_0 a_k (B_1^+ \alpha_i^{\text{nom}}, T_1, T_2, T_R) e^{-t_{ijk}/T_2} e^{-R'_2|t_{ijk} + kT_R|} e^{i(\phi_0 + \Delta\omega(t_{ijk} + kT_R))} \quad (8)$$

with $T_2^* = 1/(1/T_2 + R'_2)$. For $k \geq 0$ the magnitude decays at $T_2^{-1} + R'_2$; for $k < 0$ and $t < |k|T_R$ at $T_2^{-1} - R'_2$ (a spin echo forming at $t = |k|T_R$).

Remark 1 (Relation to the published equations). Eq. 1 of [1], $S = F^+ e^{-(R_2 \pm R'_2)T_E}$ (+ for $k \geq 0$), has the same echo-time dependence as (8); its intercept is $F_k^+ = a_k e^{-R'_2|k|T_R}$. The extra factor depends only on $|k|$, so every relation between the $\pm k$ states used in [1] (their Eqs. 4–13), and the mixing factor, are unchanged. We verified numerically that their Eqs. 7, 16 and 17 hold exactly on the steady states of (6) (Eq. 16 read as $T_1 = T_R / \ln[(Xc - 1)/(X - c)]$).

1.5 Imaging and noise

Image formation $y_{ijk} = \mathcal{M}_i \mathcal{F}(C_c S_{ijk}) + n_{ijk}$ (coil maps C_c , Fourier transform $\mathcal{F}$, sampling $\mathcal{M}_i$; scan 2 covers only the central $25\% \times 25\%$ of k_y – k_z) is linear. For the voxel-wise analysis below we model the coil-combined complex images as $S_{ijk} + n_{ijk}$ with independent circular Gaussian noise of standard deviation σ per real and imaginary part.

2 Estimation-theoretic framework

Collect the unknowns of one voxel as

$$\theta = (\log M_0, \log B_1^+, \log T_1, \log T_2, \log T_2^*, \Delta\omega, \phi_0). \quad (9)$$

For the complex Gaussian model the Fisher information is

$$\mathcal{I}(\theta) = \frac{1}{\sigma^2} \sum_{i,j,k} \text{Re} \left[\overline{\partial_\theta S_{ijk}} \partial_\theta S_{ijk}^\top \right], \quad \text{Cov}(\hat{\theta}) \succeq \mathcal{I}^{-1} \quad (10)$$

for any unbiased estimator. Because positive parameters enter through their logarithm, $\sqrt{(\mathcal{I}^{-1})_{ll}}$ is a *relative* standard deviation. With σ expressed in units of M_0 every bound scales as $\sigma/M_0 = 1/\text{SNR}(M_0)$; we report bounds *per unit* σ/M_0 . The brightest MPME image has $|S| \approx 0.088 M_0$, so the image SNR is about $0.088 \text{SNR}(M_0)$.

We use three diagnostics of conditioning:

- the *normalised information* $\tilde{\mathcal{I}} = D^{-1/2} \mathcal{I} D^{-1/2}$, $D = \text{diag } \mathcal{I}$, whose smallest eigenvalue and eigenvector identify the worst-determined combination independent of units, and its condition number κ ;
- the *variance inflation* $(\mathcal{I}^{-1})_{ll} \mathcal{I}_{ll} \geq 1$, the factor by which the bound on θ_l grows because the other parameters are unknown;
- the *fixed-parameter bias*: if θ_m is held at a wrong value $\theta_m + \delta$ and the rest is fitted, the estimates move by $\delta\theta_r = -\mathcal{I}_{rr}^{-1} \mathcal{I}_{rm} \delta$ to first order.

Jacobians are obtained by automatic differentiation through (6)–(8) and checked against finite differences.

3 The fundamental limitation: a scaling invariance

Proposition 2 (Small-angle scaling invariance). *Let $\rho = T_R/T_1$. For $\alpha \rightarrow 0$ and $\rho \rightarrow 0$ with α^2/ρ fixed, the pathway amplitudes satisfy*

$$M_0 a_k(\alpha, T_1) = \frac{M_0}{\kappa} a_k(\kappa\alpha, T_1/\kappa^2) (1 + O(\alpha^2, \rho)) \quad \text{for all } \kappa > 0, \quad (11)$$

with T_2 (hence E_2) and T_R unchanged. Consequently, through (1), the signal (8) of every scan with a small flip angle is invariant to leading order under

$$(M_0, B_1^+, T_1) \mapsto (M_0/\kappa, \kappa B_1^+, T_1/\kappa^2). \quad (12)$$

Proof for $k = 0, -1$. To leading order, $E_1 - \cos \alpha \simeq \alpha^2/2 - \rho$, $1 - E_1 \cos \alpha \simeq \rho + \alpha^2/2$, $1 - E_1 \simeq \rho$, $1 + \cos \alpha \simeq 2$, hence $p \simeq (\rho + \alpha^2/2) - E_2^2(\alpha^2/2 - \rho)$ and $q \simeq 2E_2\rho$. All of these are homogeneous of degree one in (ρ, α^2) , so the bracketed terms in (7) depend only on α^2/ρ , which is invariant under $\alpha \rightarrow \kappa\alpha$, $\rho \rightarrow \kappa^2\rho$. The remaining prefactor $\tan(\alpha/2) \simeq \alpha/2$ scales by κ and is compensated by $M_0 \rightarrow M_0/\kappa$. The corrections are relative $O(\alpha^2)$ (from $\tan, \cos$) and $O(\rho)$ (from E_1). $\square$

For higher pathways we verify the invariance numerically on the full model with all three pathways (Section 4): fixing B_1^+ wrong by ε shifts the best fit by $\delta \log T_1 = -2.06 \varepsilon$ and $\delta \log M_0 = -1.02 \varepsilon$, against the predicted -2 and -1 .

Corollary 1 (Near-singular information). *Let $v = (-1, +1, -2, 0, 0, 0, 0)$ in the coordinates of θ . For a protocol with small flip angles only, $\partial_v S_{ijk} = O(\alpha^2, \rho) \|\partial_\theta S\|$, so $v^\top \mathcal{I} v$ is small compared with the diagonal of $\mathcal{I}$, $\mathcal{I}$ is ill-conditioned, and the bounds on M_0 , B_1^+ , T_1 grow without limit as $\alpha_i^{\text{nom}} \rightarrow 0$. Acquiring several small flip angles does not help, since all of them scale with the same κ .*

Remark 2. Corollary 1 is a statement about the information contained in the data, i.e. about the forward model and the protocol. It holds for every estimator: analytic, maximum likelihood, dictionary or neural network.

3.1 How the published protocol escapes

Proposition 3 (Periodicity). $|a_k(\alpha)|$ is even and 2π -periodic in α , and $a_k(-\alpha) = -a_k(\alpha)$ for all k (a global sign). Hence, with B_1^+ known, the protocols $15^\circ/330^\circ$ and $15^\circ/30^\circ$ have identical Fisher information for every parameter.

Proof. $R_x(-\alpha) = R_z(\pi)R_x(\alpha)R_z(-\pi)$, i.e. RF phase $\varphi = \pi$. Since precession and relaxation commute with R_z , the steady state for phase π is $R_z(\pi)$ applied to the one for phase 0, which multiplies every F_k by -1 . Periodicity is that of R_x . A known, scan-wide sign does not change $\mathcal{I}$. $\square$

Proposition 3 shows that the 330° pulse brings no new information about T_1 when B_1^+ is known. Its value lies entirely in the B_1^+ derivative. The effective angle of scan 2 is $\alpha_2^{\text{eff}} = 330^\circ B_1^+ - 360^\circ \approx -30^\circ$, so

$$\frac{\partial \log |\alpha_2^{\text{eff}}|}{\partial \log B_1^+} = \frac{330}{330 - 360} = -11, \quad \text{whereas} \quad \frac{\partial \log \alpha_1}{\partial \log B_1^+} = +1. \quad (13)$$

The invariance (12) requires all angles to scale by the same factor; under a B_1^+ change the second angle moves eleven times faster and in the opposite direction, so the null direction v is lifted. The near- 360° pulse acts as a built-in B_1^+ measurement.

Remark 3 (Global ambiguity). Proposition 3 also implies that magnitudes cannot distinguish $\alpha_2^{\text{eff}} = 360^\circ \mp x$. The sign of the scan-2 FID relative to scan 1 resolves this branch (as in [1]), except at $\alpha_2^{\text{eff}} = 360^\circ$ exactly, where that signal vanishes. This is a global identifiability issue that a local bound cannot see; locally, B_1^+ is determined *best* near 360° (Fig. 4).

4 Cramér–Rao bounds and conditioning

Unless stated otherwise the tissue is the reference used for protocol optimisation in [1]: $T_1/T_2/T_2^* = 1500/70/60$ ms, $B_1^+ = 1$.

Table 1: CRLB per unit σ/M_0 : relative standard deviation for $M_0, B_1^+, T_1, T_2, T_2^*$; rad/ms for $\Delta\omega$.

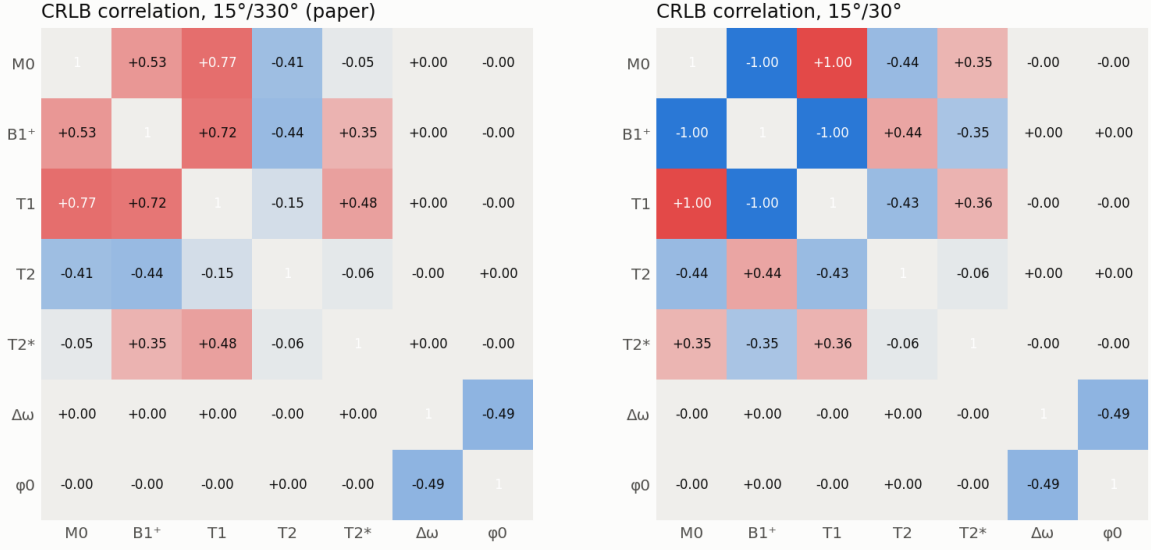
Protocol	Tissue ($T_1/T_2/T_2^*$, ms)	M_0	B_1^+	T_1	T_2	T_2^*	$\Delta\omega$
$15^\circ/330^\circ$	WM (850/66/50)	12.5	2.21	27.3	16.0	27.2	0.30
$15^\circ/330^\circ$	GM (1350/90/60)	14.7	2.75	33.1	20.7	38.2	0.31
$15^\circ/330^\circ$	reference (1500/70/60)	19.0	2.91	40.5	23.1	47.3	0.37
$15^\circ/330^\circ$	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37
$15^\circ/30^\circ$	reference	1050	1020	2110	23.1	47.3	0.37
$15^\circ/30^\circ$	reference, B_1^+ known	16.1	–	28.0	20.7	44.2	0.37

Table 1 shows the two effects at once. With $15^\circ/30^\circ$ the M_0 – B_1^+ – T_1 block is essentially singular while T_2, T_2^* and $\Delta\omega$ are untouched; with B_1^+ known the two protocols coincide (Proposition 3).

Where the remaining T_1 uncertainty comes from. Freeing parameters one group at a time gives the hierarchy

Table 2: Conditioning of the information matrix (reference tissue).

	15°/330°	15°/30°
Eigenvalues of $\tilde{\mathcal{I}}$	0.024, 0.11, 0.22, 0.51, 0.52, 1.49, 4.13	7.7×10^{-6} , 0.11, 0.29, 0.51, 0.76, 1.49, 3.84
Condition number $\kappa(\tilde{\mathcal{I}})$	173	5.0×10^5
Variance inflation $M_0/B_1^+/T_1$	16.7/4.6/26.6	51 000/7 700/72 000
Correlations (B_1^+, T_1) , (B_1^+, M_0) , (T_1, M_0)	0.72, 0.53, 0.77	-0.9999, -0.9999, 0.9999
$\partial \log T_1 / \partial \log B_1^+$, B_1^+ fixed, all data	+10.1	-2.06
$\partial \log T_1 / \partial \log B_1^+$, B_1^+ fixed, scan 1 only	-2.02	-2.02


 Figure 1: Correlation matrices of the CRLB covariance $\mathcal{I}^{-1}$. Right: the M_0 - B_1^+ - T_1 block of the small-angle protocol is perfectly correlated, the signature of the invariance (12).

Unknown parameters	T_1 bound (per unit σ/M_0)
T_1 only	7.9
+ T_2 , T_2^* , $\Delta\omega$, ϕ_0	21.1
+ M_0	28.0
+ B_1^+ (15°/330°)	40.5
+ B_1^+ (15°/30°)	2110

With the published protocol B_1^+ costs a factor 1.45. The larger costs are the transverse relaxation times (2.7) and M_0 (1.3): at 15° and $T_R = 25$ ms ($T_1 \gg T_R$) T_1 is weakly encoded in the steady-state amplitudes, and a longer T_1 lowers all amplitudes in a way a larger M_0 largely compensates.

Protocol dependence. Fig. 3 shows that the T_1 bound with B_1^+ unknown falls steeply with α_2^{nom} and, from about 210°, stays within $1.5\times$ of the B_1^+ -known bound. Over (α_1, α_2) the optimum is 10°/340° (bound 32.9), only 19% better than 15°/330° (Fig. 5). Across a 3 T-typical B_1^+ range 0.7–1.3 the T_1 bound stays between 40 and 65, and it grows with T_1 itself (15 at 300 ms, 100 at 4000 ms; Fig. 4). Removing the $k = +1$ pathway raises the T_1 bound from 40.5 to 67; removing the echo pathway $k = -1$ raises it to 146 and the T_2 bound from 23 to 208.

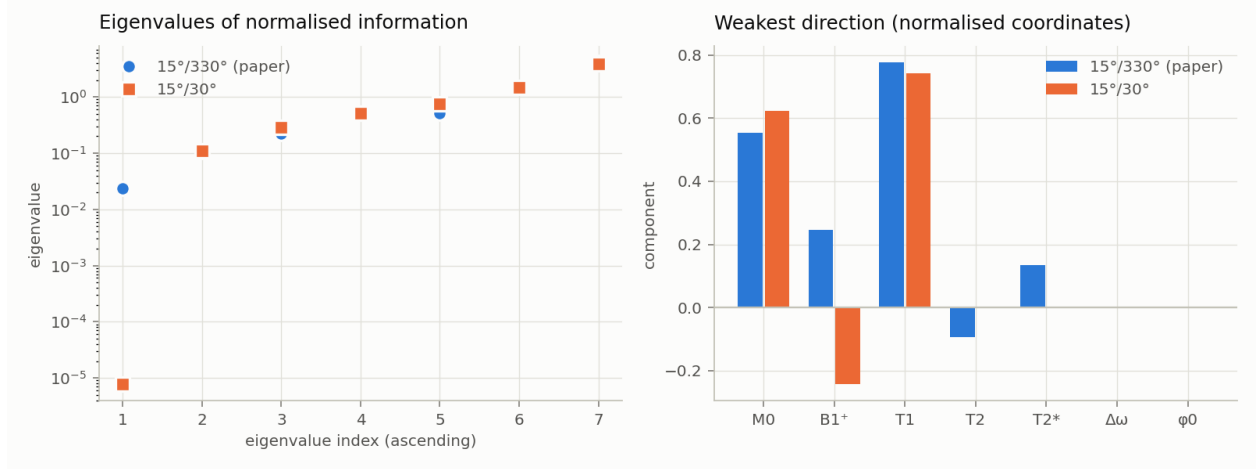


Figure 2: Left: eigenvalues of the normalised information. Right: the weakest eigenvector. For 15°/30° it is the (normalised) symmetry direction; for 15°/330° the smallest eigenvalue is three thousand times larger and the weakest direction is mostly M_0 and T_1 moving together.

5 Forward model or inverse?

The CRLB characterises the forward model and protocol; it says nothing about whether a particular reconstruction attains it. To separate the two we compare three estimators of the *same* model on simulated data:

1. the **analytic inverse** of [1], implemented equation by equation: $\Delta\omega$ from echo phases; R_2 , R'_2 from a log-linear fit of their Eq. 1 to scan 1; real-valued F states (Eq. 18); the mixing factor (Eqs. 10–11); the flip angle from Eq. 15; T_1 and M_0 from Eqs. 16–17 using scan 1. We also run it with B_1^+ given, mimicking the paper’s low-resolution, polynomial-smoothed B_1^+ map;
2. a **model-fit baseline**: the same sequential structure, but with B_1^+ , T_1 , M_0 fitted by least squares to the FID and echo amplitudes of both scans;
3. a **joint maximum-likelihood (ML) fit** of (8) to all complex data (Section 5.1).

All three are exact on noise-free data (relative errors below 10^{-11} for the analytic inverse). Table 3 reports robust standard deviations (interquartile range/1.349 of the log-estimates) over 1000 noise realisations.

5.1 The joint maximum-likelihood estimator

Objective. Stack the real and imaginary parts of all measured voxel signals ($2 \text{ scans} \times 3 \text{ pathways} \times 3 \text{ echoes}$) into $y \in \mathbb{R}^{36}$ and the model (8) into $s(\theta) \in \mathbb{R}^{36}$, with θ as in Section 2. Under independent Gaussian noise of variance σ^2 per component the negative log-likelihood is, up to constants,

$$-\log p(y \mid \theta) = \frac{1}{2\sigma^2} \|r(\theta)\|^2, \quad r(\theta) = s(\theta) - y, \quad (14)$$

so the ML estimate is the nonlinear least-squares solution $\hat{\theta} = \arg \min_{\theta} \|r(\theta)\|^2$. It does not depend on σ . The model $s(\theta)$ is evaluated with the exact steady-state solver (6) (256 isochromats), so the

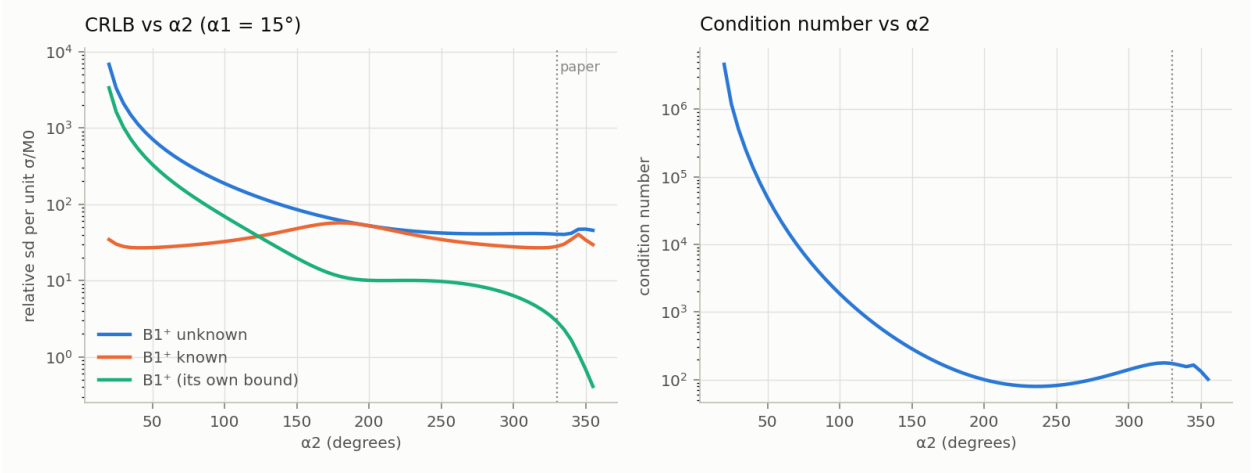


Figure 3: Bounds and condition number versus the nominal second flip angle ($\alpha_1^{\text{nom}} = 15^\circ$).

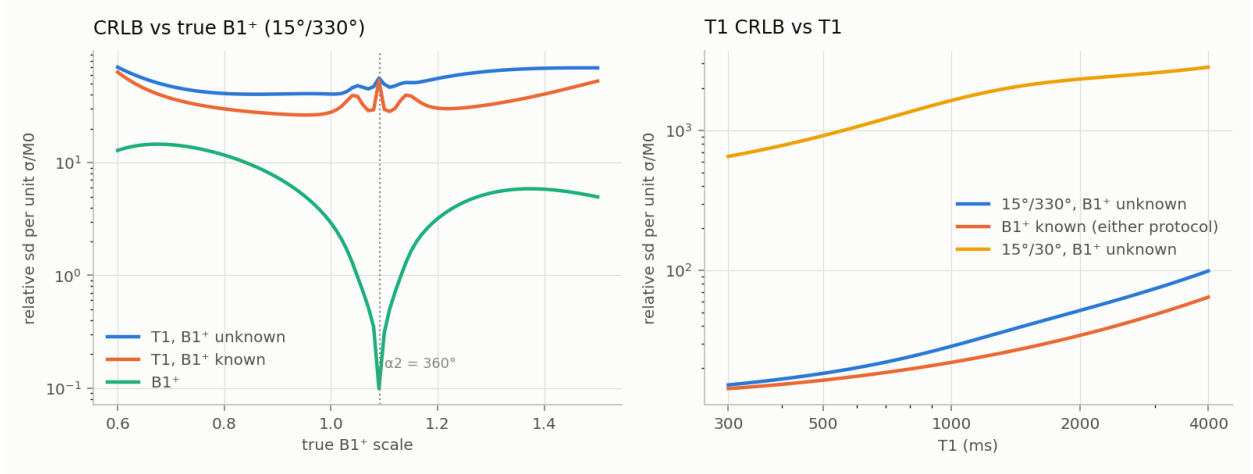


Figure 4: Left: bounds versus the true B_1^+ for the published protocol; B_1^+ is best determined where $\alpha_2^{\text{eff}} = 360^\circ$. Right: T_1 bound versus T_1 .

estimator uses the same forward model as the CRLB, including all three pathways, both scans, and the phase.

Parameterisation. The five positive parameters are optimised in log coordinates, which keeps them positive without constraints and makes the problem better scaled (all log-parameters have comparable sensitivities). $\Delta\omega$ and ϕ_0 are unconstrained. The only bound is $\log B_1^+ \in [-1, 1]$ ($B_1^+ \in [0.37, 2.7]$), a safeguard against divergent steps far outside any physical range.

Initialisation. The magnitude parameters and $\Delta\omega$ are taken from a sequential estimator: the model-fit baseline, or the analytic inverse where its output is valid. The global phase, which the sequential estimators do not provide, has a closed-form optimum for fixed remaining parameters.

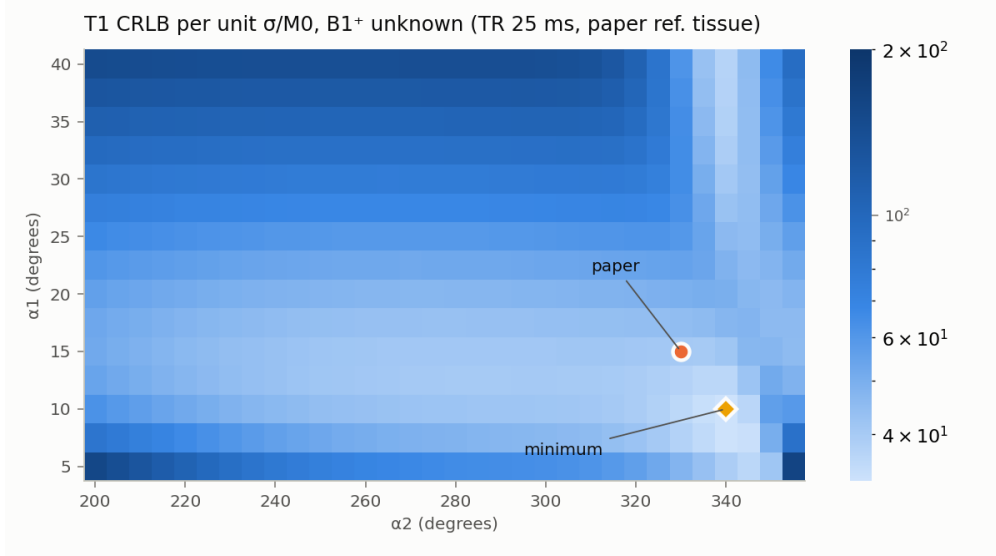


Figure 5: T_1 bound with B_1^+ unknown over nominal (α_1, α_2) .

With $S_{ijk}^{(0)}$ the model evaluated at the initial values and $\phi_0 = 0$,

$$\phi_0^{(0)} = \arg\left(\sum_{i,j,k} S_{ijk} \overline{S_{ijk}^{(0)}}\right), \quad (15)$$

which minimises $\|r\|^2$ over ϕ_0 alone.

Levenberg–Marquardt iteration. Each voxel is fitted independently; all voxels are processed in parallel as a batch. At iteration n , with $J = \partial r / \partial \theta \in \mathbb{R}^{36 \times 7}$:

1. *Jacobian.* Forward differences, $J_{:,l} = [r(\theta + h e_l) - r(\theta)]/h$ with $h = 10^{-6}$ (all quantities in double precision).
2. *Damped Gauss–Newton step* with Marquardt’s diagonal scaling,

$$(J^T J + \lambda \text{diag}(J^T J)) \Delta = -J^T r, \quad \theta_{\text{new}} = \theta + \Delta. \quad (16)$$

The diagonal scaling makes the step invariant to rescaling individual parameters, which matters here because the information matrix spans several orders of magnitude.

3. *Acceptance.* If $\|r(\theta_{\text{new}})\|^2 < \|r(\theta)\|^2$ the step is accepted and $\lambda \leftarrow 0.3\lambda$ (towards Gauss–Newton); otherwise it is rejected and $\lambda \leftarrow 10\lambda$ (towards scaled gradient descent). λ starts at 10^{-2} and is kept in $[10^{-9}, 10^9]$.

We run a fixed 40 iterations. Near the optimum $\lambda \rightarrow 0$ and the iteration becomes Gauss–Newton, which converges quadratically for small-residual problems such as this one; in noise-free tests the parameters are recovered to 10^{-15} relative error from starts perturbed by 5%.

Multiple starts. Nonlinear least squares can converge to a local minimum, in particular across the 360° branch of the second flip angle or, for the small-angle protocol, anywhere along the nearly flat symmetry valley (12). We therefore run the iteration from two starts and keep, per voxel, the

solution with the lower final cost. For $15^\circ/330^\circ$ the starts are the model-fit and the analytic-inverse estimates. For $15^\circ/30^\circ$ the second start is the first moved along the symmetry direction ($B_1^+ \times 1.1$, $T_1/1.21$, $M_0/1.1$).

Why the ML fit should attain the bound. At the solution, the Gauss–Newton approximation of the covariance is $\sigma^2(J^\top J)^{-1}$, and $J^\top J/\sigma^2$ evaluated at the true parameters is exactly the Fisher information of Section 2. Under standard regularity conditions the ML estimator is consistent and asymptotically efficient: as $\text{SNR} \rightarrow \infty$ its covariance converges to $\mathcal{I}^{-1}$ and its bias vanishes. At finite SNR it can deviate from the bound when the log-likelihood is far from quadratic over the noise-induced spread, which is exactly what happens along a nearly flat direction. Agreement between the ML spread and the CRLB at a given SNR therefore shows both that the bound is attainable *and* that the problem is locally well posed at that SNR. For the small-angle protocol the ML fit still tracks its bound, but that bound is itself of order 100%: there an efficient estimator is useless, which is the practical face of Corollary 1.

Cost. Each iteration needs eight model evaluations (one plus seven finite-difference directions). For 10^3 voxels the full fit takes about a minute on a laptop CPU. That is fast enough for simulation studies, but slow for whole-brain data ($\sim 5 \times 10^6$ voxels), which is one motivation for the learned estimators in the project plan.

Table 3: Monte Carlo robust standard deviation (%) of the log-estimates, 1000 noise realisations, reference tissue, $B_1^+ = 1$. SNR is relative to M_0 (peak image $\text{SNR} \approx 0.088\times$). Last column: fraction of voxels with undefined T_1 .

Protocol	SNR(M_0)	Estimator	T_1	B_1^+	T_2	M_0	T_1 undefined
$15^\circ/330^\circ$	300	CRLB	13.5	1.0	7.7	6.3	–
		Joint ML	13.7	0.9	7.5	6.0	0.0%
		Model fit	90.7	3.7	36.8	56.9	0.0%
		Analytic inverse [1]	102.6	46.1	37.0	57.1	1.9%
		Analytic inverse, B_1^+ given	35.9	–	–	23.3	1.9%
$15^\circ/330^\circ$	1000	CRLB	4.1	0.3	2.3	1.9	–
		Joint ML	3.9	0.3	2.5	1.9	0.0%
		Model fit	11.4	1.3	11.0	8.5	0.0%
		Analytic inverse [1]	12.7	1.5	11.2	7.6	0.0%
		Analytic inverse, B_1^+ given	10.7	–	–	6.5	0.0%
$15^\circ/330^\circ$	3000	CRLB	1.4	0.1	0.8	0.6	–
		Joint ML	1.3	0.1	0.8	0.6	0.0%
		Model fit	4.0	0.4	3.8	2.9	0.0%
		Analytic inverse [1]	4.4	0.4	3.7	2.6	0.0%
		Analytic inverse, B_1^+ given	3.9	–	–	2.2	0.0%
$15^\circ/30^\circ$	1000	CRLB	210.6	102.3	2.3	104.8	–
		Joint ML	211.7	103.7	2.2	105.3	0.0%
		Model fit	287.6	138.0	11.0	149.0	0.0%
$15^\circ/30^\circ$	3000	CRLB	70.2	34.1	0.8	34.9	–
		Joint ML	77.1	37.3	0.7	38.1	0.0%
		Model fit	225.6	109.7	3.8	113.9	0.0%

Reading Table 3.

- **The bound is attainable.** For $15^\circ/330^\circ$ the joint ML fit matches the CRLB for T_1 , B_1^+ , T_2 and M_0 at every SNR tested, to within the Monte Carlo error (T_1 : 13.7 vs 13.5%, 3.9 vs 4.1%, 1.3 vs 1.35%). The forward model therefore *contains* the information that Table 1 promises, and it can be extracted already at $\text{SNR}(M_0) = 300$; the published protocol is not intrinsically unable to separate B_1^+ and T_1 .
- **The analytic inverse loses a factor of about three** at high SNR (T_1 : 3.1–3.3, B_1^+ : 4.5–5.1, T_2 : 4.8). At $\text{SNR}(M_0) = 300$ (peak image SNR ≈ 26) the T_1 spread is $7.6\times$ the bound, the B_1^+ spread $47\times$, and 1.9% of voxels have no valid T_1 at all: Eq. 16 requires $(Xc - 1)/(X - c) > 1$, which noise violates. Supplying a noise-free B_1^+ helps but does not close the gap ($2.7\times$ the bound at SNR 300 and SNR 1000), because the dominant loss is in the mixing factor X , built from extrapolated and differenced F -state estimates. The model-fit baseline, which shares the sequential structure, behaves similarly: the loss comes from the sequential structure, not from the specific closed forms.
- **The small-angle control fails for every estimator.** For $15^\circ/30^\circ$ the ML fit is again efficient (T_1 : 212 vs 211% at SNR 1000) but the bound itself is useless: T_1 and B_1^+ spreads are of order unity in log units even at SNR 3000, while T_2 is unaffected. Here the limitation is the forward model and protocol (Corollary 1), and no reconstruction method can remove it.

Sources of the estimator-specific loss. The analytic inverse (i) is *sequential*: R_2 and R'_2 are fixed from scan 1 before the flip angle and T_1 are estimated, so their errors propagate instead of being traded off jointly; (ii) uses *closed-form nonlinear transformations* (differences of extrapolated states, ratios, logarithms) whose noise amplification is not controlled and whose domain can be left by noise; (iii) uses only *scan 1* for T_1 , with B_1^+ held fixed, which raises the per-voxel bound from 40.5 to 57 and turns a B_1^+ error ε into a T_1 bias -2.02ε (Table 2); and (iv) discards phase (Eq. 18), which costs little here because $\Delta\omega$ and ϕ_0 decouple from the other parameters (Fig. 1). Point (iii) is partly a design choice of the acquisition: scan 2 is acquired only at low resolution, so the full-resolution joint bound of 40.5 is not available to any estimator in that design. At the low resolution both scans have about four times the per-voxel SNR, giving a B_1^+ bound of $2.91/4 \approx 0.73$ per low-resolution voxel before spatial smoothing.

6 Implications

1. The published protocol is sound from an information standpoint: the near- 360° pulse removes the invariance that makes B_1^+/T_1 separation impossible with small angles.
2. No voxel-wise estimator, learned or not, can beat Table 1. The realistic voxel-wise gain over the analytic inverse is the factor ≈ 3 that the ML fit recovers.
3. Larger gains require prior information across voxels. B_1^+ is spatially smooth, so treating it as nearly known removes its factor 1.45; the remaining confounding with M_0 and the transverse relaxation times can only be reduced by image priors.
4. When B_1^+ is supplied to a joint estimator that also uses the high-flip data, a B_1^+ error is amplified about tenfold into T_1 (Table 2); B_1^+ should be estimated jointly or the high-flip data excluded from the T_1 step.

Table 4: Origin of each difficulty.

Difficulty	Origin	Remedy
M_0 - B_1^+ - T_1 invariance (12) for small flip angles	forward model + protocol	a pulse whose effective angle does not scale with B_1^+ (e.g. near 360° , as published)
Residual T_1 bound (40.5 per unit σ/M_0): confounding with T_2 , T_2^* , M_0 , B_1^+	forward model + protocol	SNR, averaging, spatial priors; protocol tuning ($\leq 20\%$)
Branch ambiguity $360^\circ \mp x$	forward model (periodicity)	phase of the scan-2 FID
Factor ≈ 3 above the bound; undefined T_1 at moderate SNR	analytic inverse	joint (ML or learned) estimation
Bound 57 instead of 40.5 per voxel; T_1 bias -2ϵ per B_1^+ error ϵ	two-stage design (low-resolution scan 2, scan-1-only T_1)	joint estimation with a smooth- B_1^+ prior
In-vivo calibration $\beta = 1.24$ (Eq. 19 of [1])	model mismatch (not variance)	richer forward model (finite pulses, MT, ...)

Caveats. The CRLB is local and assumes an exact model, Gaussian noise and unbiased estimators. The model uses instantaneous RF pulses (no off-resonance nutation, Eq. 8 of [1]), a Lorentzian R'_2 line shape, no diffusion and no magnetisation transfer. The time from the pulse to the first echo is assumed. Except where stated, both scans are treated at full resolution.

A Reproducibility

Forward model	<code>src/mpme/epg.py</code> , <code>signal.py</code> , <code>sequence.py</code>
Analytic inverse of [1]	<code>src/mpme/paper.py</code>
Model-fit baseline	<code>src/mpme/analytic.py</code>
Joint ML fit	<code>src/mpme/mle.py</code>
Fisher information, CRLB	<code>src/mpme/crlb.py</code>
Tables 1–2, figures	<code>python scripts/crlb_analysis.py</code>
Table 3	<code>python scripts/estimator_efficiency.py</code>

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